# Pupillary Pathway

Figures used: 14.11, 14.12, 14.13, 14.3, 14.16, 14.19, 14.17, 14.18, 14.21

I. Pupillary pathway

A. Afferent pathway follows the visual fibers to the optic tract

1. The pupillary fibers leave the optic tract and travel within the brachium of the superior colliculus to the pretectal area of the midbrain.

B. Central pathway: fibers travel from the pretectal nucleus to the Edinger Westfall preganglionic cells

C. Efferent pathway: follows cranial nerve 3

D. Clinical correlate: Direct and consensual pupillary light response

1. Direct pupillary light response:

2. Consensual pupillary light response:

II. Disruption of the afferent pathway

A. Disruption of the afferent pathway will affect the direct response of the ipsilateral eye and the consensual response of the contralateral eye.

B. Swinging flashlight test

1. The patient fixates on a distance target, and the doctor moves a light from one eye to the other several times.

2. It is important to make sure each pupil is illuminated with an equal amount of light.

3. If there is no damage to the visual pathway, there will be little or no change in pupil size when switching the light from one eye to the other.

4. If there is damage to the afferent pathway, both pupils will dilate quicker when shining light into the pupil on the side ipsilateral to the damage.

C. Cause of RAPD:

1. Retina

2. Optic nerve

3. Optic tract

4. Superior brachium

5. Clinical correlate:  media opacities do not cause an ipsilateral RAPD

III. Disruption in the central nervous system pathway

A. Lesions of the midbrain can affect the fibers between the pretectal and Edinger-Westphal preganglionic cells

B. Light-near dissociation

1. Near pupil response bypasses the central portion of the pupillary light response

2. Near pupil response pathway

a. Afferent

b. CNS

c. Efferent

3. Clinical correlate:  Argyll Robertson pupil
    a. Bilateral light near dissociation
    b. Without any light stimulus, the pupils will be very small.
    c. Most commonly due to neurosyphilis
4. Clinical correlate:  dorsal midbrain syndrome
    a. Most commonly due to a pineal gland abnormality
    b. Bilateral light near dissociation
    c. Pupils are normal to mid-dilated

IV. Disruption in the efferent parasympathetic pathway

A. A parasympathetic lesion will cause the ipsilateral eye to show poor direct, consensual, and near pupillary responses.

1. Ipsilateral mydriasis

2. Anisocoria worse in bright light

B. Causes

1. Cranial nerve 3 palsy

2. Tonic pupil

a. Caused by damage to the ciliary ganglion or short ciliary nerves

(1) Autonomic dysfunction

(2) Orbital mass, trauma, or surgery

(3) Idiopathic (Adie tonic pupil)

b. Segmental constriction.

c. Decreased corneal sensitivity

d. Cholinergic denervation supersensitivity

3. Pharmacologically dilated pupil

a. Pharmacologic testing: 1% pilocarpine

V. Disruption of the efferent sympathetic pathway: Horner syndrome

A. Miosis

B. Anisocoria greater in dim light

C. Normal reaction to light

D. Dilation lag

E. Ptosis

F. Facial anhidrosis (absence of sweating)

G. Pathway

1. Central neuron (first order neuron)
    a. Hypothalamus → brainstem → cervical spinal cord

2. Preganglionic neuron (second order neuron)
    a. Cervical spinal cord → apex of the lung → superior cervical ganglion

3. Postganglionic neuron (third order neuron)
    a. Internal carotid artery → cavernous sinus → nasociliary nerve → long ciliary nerves → iris dilator

H. Pharmacologic testing

1. Apraclonidine 0.5 or 1%

    a. Alpha adrenergic agonist

VI. Physiologic anisocoria